

Exercise 1: Mobility price data

Retrieve the average prices of intercity bus connections between the “TU9” university cities (Aachen, Berlin, Braunschweig, Darmstadt, Dresden, Hannover, Karlsruhe, München, Stuttgart) from the Flixbus website (<https://global.flixbus.com/>). Use a lead time of ~45 days.

As a next step, focus on a single connection (e.g., Berlin → Hamburg) and vary the lead time (i.e., how much in advance you book the trip; e.g. from 0 to 100 days). How does the price change for rides further in the future? Are there any day-of-week effects?

The results might look something like this:

	AC	B	BS	DA	DD	HA	KA	M	S
Aachen		37.98	34.66	18.48	34.66	20.48	22.84	35.38	27.70
Berlin	34.55		20.99	29.80	20.99	13.81	39.26	25.43	28.42
Braunschweig	33.99	24.10		28.48		12.98	40.61	40.06	43.19
Darmstadt	21.48	40.99	38.98		38.98	31.49	10.60	22.63	9.34
Dresden	33.99	24.10		28.48		12.98	40.61	40.06	43.19
Hannover	21.99	15.03		30.90			38.91	34.76	34.88
Karlsruhe	23.99	49.90	43.73	15.49	43.73	49.53		16.04	6.53
München	35.23	34.89	44.78	21.13	44.78	45.12	15.61		11.05
Stuttgart	29.30	41.96	45.40	9.49	45.40	45.80	5.70	11.47	

